



Ghulam Ishaq Khan Institute (GIKI)

Project

Subject: ICT	Course Code: CS202
Class: BS CS 3rd	Submission Deadline: 21/Dec/2025
Instructor: M Talha Ashfaq	Total Marks: 100

Note (Read notes & instructions first)

CHEATING/COPY CASE or **LATE SUBMISSION** even 1 minute late will be graded as **STRAIGHT ZERO MARKS**.
So be on time make no excuse.

(Read Carefully)

Dataset Preprocessing and Visualization in R

Objective:

This assignment aims to help students understand the fundamental steps in dataset preprocessing and data visualization. Students will select a dataset, clean and preprocess the data, explore it using visualizations, and provide insights. This will help develop skills in data manipulation, cleaning, and visualization — essential tasks in data analysis and machine learning.

Dataset Selection:

Choose any dataset that has both numerical and categorical variables (e.g., from Kaggle, UCI, or any public dataset repository).

You may consider datasets such as:

- **Iris dataset** (classification of iris species based on flower features).
- **Titanic dataset** (predict survival based on passenger features).
- **Wine dataset** (classification of wine quality).
- **Breast Cancer dataset** (diagnosis of cancerous or non-cancerous samples).

Note: Ensure the dataset has at least 3 features and a target variable for classification or regression tasks.

Steps to Follow:

1. Data Importing and Preprocessing

Load the Dataset: Import the dataset into R using `read.csv()` or `readr::read_csv()`.

Data Cleaning:

- **Missing Values:** Identify and handle missing data by either removing rows/columns or imputing missing values (e.g., using the mean or median for numerical data).
- **Duplicate Rows:** Check for and remove any duplicate rows using `duplicated()`.
- **Categorical Encoding:** Convert categorical variables into factors, if necessary.

2. Exploratory Data Analysis (EDA)

Summary Statistics: Display summary statistics of the numerical features (e.g., mean, median, standard deviation, etc.).

Target Variable Analysis: Examine the distribution of the target variable.

- If the target variable is categorical (e.g., **yes/no**, **class1/class2**), visualize the count distribution.

Feature Distribution:

- For numerical features, plot histograms or box plots to understand their distribution.
- For categorical features, use bar plots to show the frequency of each category.

3. Data Visualization

Correlation Matrix: For numerical features, plot a correlation matrix to understand relationships between variables.

Scatter Plots: Visualize relationships between pairs of numerical features.

Boxplots for Target vs. Feature: For numerical features, create boxplots to compare the distribution of the feature across different classes of the target variable.

4. Feature Engineering

Feature Scaling: If your data contains features with very different scales, consider normalizing or standardizing the features (especially important for models like k-NN).

Create New Features: Based on your understanding of the dataset, you may try creating new features that could help improve model performance (e.g., ratios, combinations of existing features, etc.).

5. Model Preparation

Split Data into Train and Test Sets: Split the data into training and testing sets (usually an 80/20 or 70/30 split).

6. Documentation and Reporting

Write a brief report that explains your findings and insights from the EDA.

In your report, include the following:

- **Dataset Description:** Provide the name of the dataset, source, and a brief overview of the features and target variable.
- **Key Findings from EDA:**
 - Summary statistics of the numerical features.
 - Distributions of categorical and numerical features.
 - Insights drawn from the correlation matrix or visualizations.
- **Data Cleaning:** Describe how you handled missing values and duplicates, if any.
- **Visualizations:** Present the most important visualizations that highlight the key patterns in the data.
- **Next Steps:** Suggest possible further analyses, improvements, or models that could be applied.

Deliverables:

Code: Submit your R script or R Markdown file that includes:

- Data import and preprocessing steps.
- Visualizations (e.g., histograms, box plots, scatter plots).
- Feature engineering (if done).
- Any necessary comments to explain your code.

Visualizations: Include relevant visualizations that help explain the data (e.g., plots showing distributions, relationships between features, etc.).

Report: Submit a brief report (max 2 pages) with:

- Dataset description.
- Key findings from the EDA.
- Insights from the visualizations.

- Any additional steps for further analysis.

Grading Criteria:

- **Code Quality (30%):** The code should be clear, efficient, and well-commented.
- **Correctness (40%):** Proper handling of missing values, duplicates, and correct usage of visualization techniques.
- **Analysis (20%):** A thorough and insightful EDA that includes meaningful visualizations and interpretations.
- **Creativity (10%):** Innovative approaches to data exploration, feature creation, or visualization.